SYS_DOMAIN: ARTIFICIAL INTELLIGENCE & NEURAL NETWORKS (AINN) / CYBER CTI

GHOST PROTOCOL — MASTER IMPLEMENTATION SPEC

Autonomous AI Agent, Neural ML Pipeline, Chimera UI & HIBP Breach Intelligence Engine

1. EXECUTIVE OVERVIEW & SUBJECT DOMAIN (AINN)

GHOST PROTOCOL is an autonomous cyber-threat observation, neural classification, graph correlation, and explainable AI platform. Academically positioned under the subject **Artificial Intelligence and Neural Networks (AINN)**, it applies multi-modal deep learning, natural language processing (NLP), and temporal graph memory to Open Source Intelligence (OSINT) and dark web observation. This master implementation specification defines the complete engineering roadmap for Antigravity to execute against the existing DarkWatch repository and Docker laboratory.

Core Architectural Constraints & Guarantees:

• **Frozen Machine Learning Core:** Zero retraining or modifications to HackerSignal DistilBERT (8 exploit classes, 91.30% test acc), XGBoost (3 transport modes), Temporal Autoencoder v2 (14 session features -> 8-dim latent z), and HM-TIF Fusion.

• **Whonix Fail-Closed Isolation:** All external traffic is forcibly routed through Whonix SOCKS5 proxy (socks5h://10.152.152.10:9050). Direct connections fail closed instantly.

• **Constrained Chimera UI Palette:** Strict 4-color palette: Deep Black/Slate (#0b0f19 / #111827), Green (#059669), Sky Blue (#0284c7), and Red (#dc2626) — no unnecessary colors or high-latency CSS animations.

• **Autonomous AI Agent & Recursive Link Discovery:** Background worker loop continuously observes validated sources, extracting outbound .onion, t.me/s/, and paste links in real time.

• **Have I Been Pwned (HIBP) Breach Search Engine:** Integrated credential & identity leak lookup searching observed breach dumps, displaying breach histories, exposed fields, and provenance.

2. FROZEN NEURAL ML & FUSION PIPELINE ARCHITECTURE

The neural core processes multi-modal session telemetry across three parallel encoders before dynamic cross-attention fusion:

Neural / ML Model	Input Domain	Architecture & Vector Output	Role in GHOST Protocol
HackerSignal DistilBERT	Sanitizer-cleaned text	6-layer DistilBERT, 768-dim [CLS] vector, 8 exploit classes	Classifies threat type (RCE, Injection, XSS, DoS, Auth, File Inc, Mem Corp, Info Disc)
XGBoost Network Model	80 CICFlowMeter features	Multiclass Gradient Boosted Trees, 64-dim vector	Identifies encrypted transport mode (Clearnet, Tor, VPN)
Temporal Autoencoder v2	14 session action timestamps	LSTM Autoencoder, 8-dim latent z, MSE loss	Detects session behavioral anomalies & automated bot bursts
HM-TIF Fusion Engine	Text, Net, Temp embeddings	Hierarchical Cross-Attention with missing-modality mask	Computes Dynamic Risk Score (0.00 - 1.00) with 503 barrier resilience

3. AUTONOMOUS AI AGENT & REAL-TIME OBSERVATION ENGINE

GHOST Protocol operates an autonomous background agent loop. During continuous polling of validated targets, the **Recursive Link Extractor** scans clean body text to capture outbound .onion, t.me/s/, and pastebin links in real time, inserting unindexed targets into the discovery queue.

Target Type	Observation Driver	Session & Auth Protocol	JavaScript State
Telegram Public Channels	t.me/s/SliceForLifeee HTML Preview	No login, no API keys, incremental message ID tracking	Disabled by Default
Dread Dark Web Forum	Playwright SOCKS5 Browser Context	Authorized 2-step verification -> Volatile In-Memory Cookie Vault	Enabled for Handshake; Disabled for Bulk Polling
Ransomware Extortion Portals	httpx.AsyncClient over Whonix	Passive GET requests, raw/clean SHA-256 hashing	Disabled by Default
Docker Lab Testbed	Containerized Mock Services	503 Barrier simulation, state validation (/barrier, /forum, /leak)	Configurable for Testing

4. PROJECT CHIMERA UI / SOC WORKSTATION SPECIFICATION

The SOC Workstation employs a strict 3-part layout with Project Chimera styling and a constrained 4-color palette (Black, Green, Sky Blue, Red). Zero transition delays or keyframe animations ensure immediate responsiveness.

Workstation Section	Width / Layout	Contained Features & Components	Real Data Rendered
LEFT SIDEBAR	240px Fixed Width	17 Menu Items: Dashboard, Source Discovery, Sources, Live Observation, Threats, Entities, Correlation, Infrastructure, CTI, STIX, MITRE, XAI, OSINT, Investigations, Timelines, GHOST AI, Settings	Active route highlight, system state badge
CENTER WORKSPACE	Flex: 1 Dynamic Viewport	Explained Alert Cards, Cytoscape.js STIX 2.1 Graph Canvas, Entity Profile & Alias Matrix, Interactive User Timeline, GHOST AI Chatbot Console	Dynamic risk scores, Tier 1 token highlights, SHAP bar charts, HIBP breach results
RIGHT TELEMETRY HUD	320px Fixed Width	Live Algorithm Status: DistilBERT (CUDA), XGBoost, Temporal AE v2, HM-TIF; Whonix SOCKS5 status, Cookie Vault Health, Live Risk Score Gauge, STIX/MITRE counters	Real-time SSE stream (/api/v1/telemetry/stream)

5. NATURAL LANGUAGE GHOST AI & MODEL EXPLANATION ENGINE

The GHOST AI Chatbot understands plain human language queries and translates them into structured graph and database searches. When an analyst inspects an alert or entity, the system provides an explicit, transparent breakdown of model inference:

Model Component	Classification Output	XAI / Explanation Rendered in UI
HackerSignal DistilBERT	Exploit Class (e.g., RCE, Injection, XSS, DoS)	Tier 1 Cross-Attention Token Highlights (highlighting terms like 'RCE exploit', 'CVE-2023-4966' in red)
XGBoost Network Model	Transport Class (Clearnet, Tor, VPN)	Flow feature importances (packet size variance, flow duration)
Temporal Autoencoder v2	Behavioral Anomaly Flag (Threshold 95th percentile)	MSE Reconstruction Loss graph indicating automated bot activity bursts
HM-TIF Fusion Engine	Fused Dynamic Risk Score (0.00 - 1.00)	Modality weight breakdown (Text: 0.45, Net: 0.35, Temp: 0.20) with 503 missing-text resilience flag

6. UNIFIED THREAT SEARCH, CATEGORIZATION & HIBP ENGINE

Unified Threat Categorization Engine: When an analyst searches for any indicator or term, GHOST Protocol automatically categorizes and explains the threat type: *Data Breach*, *Ransomware Extortion*, *Malware Drop*, or *Campaign Operation*.

Have I Been Pwned (HIBP) Breach Search Feature: Integrates an identity exposure lookup engine. Analysts can query emails, usernames, domains, or credentials against observed breach databases and paste dumps. Returns breach history, exposed fields (passwords, hashes, PII), exposure dates, and collection provenance.

7. ENTITY RESOLUTION & USER ACTIVITY SECTION

When an analyst clicks on a username, handle, or threat actor profile, the system displays the **User Activity Section**. Rather than treating posts as isolated events, GHOST Protocol maps the actor's historical appearance across sites and timestamps:

• **Composite Resolution Score:** S_comp = 0.40 * JaroWinkler + 0.35 * DistilBERT_Cosine + 0.25 * Graph_Jaccard
• **Activity Map:** Chronological timeline of observations, site behavior, post frequency, and alias changes.
• **Infrastructure Tracking:** Detects when an actor migrates hosts (e.g. *ShadowFox* moving from abc.onion to xyz.onion), creating a REPLACED_BY edge.

8. ANTIGRAVITY 14-PHASE IMPLEMENTATION CHECKLIST

Phase Name	Engineering Deliverable & Acceptance Criteria
Phase 1: Repository Audit	Verify existing ML imports (DistilBERT, XGBoost, Temporal AE v2, HM-TIF) and baseline unit tests.
Phase 2: Database Schema Extension	Extend annotation_store.py with sources_registry, entities, entity_aliases, graph_edges, breach_claims, and hibp_dumps tables.
Phase 3: Whonix Fail-Closed Boundary	Implement verify_whonix_gateway() socket check in whonix_collector.py enforcing socks5h://10.152.152.10:9050.
Phase 4: Source Registry Manager	Create darkwatch/discovery/source_registry.py for managing target sources, statuses, and discovery queues.
Phase 5: Telegram Public Web Observer	Create darkwatch/observer/telegram_public_observer.py targeting t.me/s/SliceForLifeee HTML preview without login.
Phase 6: Dread Session Observer	Create darkwatch/observer/dread_session_observer.py with Playwright SOCKS5 manager, 2-step verification, and volatile cookie vault.
Phase 7: Link Extraction Engine	Create darkwatch/discovery/link_extractor.py for regex extraction of outbound .onion and t.me URLs in real time.
Phase 8: Threat Filter Module	Create darkwatch/filter/threat_filter.py implementing natural-language threat pre-filtering and regex CTI tagger.
Phase 9: Pipeline Ingestion Bridge	Modify darkwatch/pipeline/lab_bridge.py to accept Stage 1 outputs and run frozen DarkWatch ML.
Phase 10: Entity Resolution & HIBP Engine	Create darkwatch/resolution/entity_resolver.py and hibp_service.py for composite match scoring and breach lookups.
Phase 11: Sherlock OSINT Engine	Create darkwatch/investigation/sherlock_wrapper.py for transient surface-web handle queries.
Phase 12: Graph Correlation Engine	Create darkwatch/cti/lookup_service.py exposing multi-hop graph queries and STIX 2.1 bundle search.
Phase 13: Chimera SOC Dashboard UI	Create darkwatch/dashboard/app.py with 3-part grid layout, Project Chimera dark theme, and telemetry HUD.
Phase 14: Docker Lab Verification	Execute 20-Scenario Docker lab test suite and verify acceptance criteria.